4-Channel Remote Igniter Controller

Design Summary v2 · June 2026 · Hammond 1455N1601BK

DESIGN CHANGES FROM V1

WAS Single **12.8V 6Ah LiFePO4 battery** for switching power + separate 18650 cell + TP4056 + MT3608 for 5V logic power (two-battery system)

NOW **2× Samsung 30Q 18650 in series (7.4V)** + 2S BMS board as the single battery pack. MT3608 boosts 7.4V → 5V for ESP32 and relay coils. Raw 7.4V goes directly to relay COM terminals for igniter firing. One battery, one boost converter.

WAS 12V relay module (designed for 12V coil voltage)

NOW **5V relay module** (VNFOCKQSH blue board) — coils run on 5V from the MT3608, simplifying power architecture to a single rail

WAS LM2596 buck converter stepping 12V → 5V

NOW **MT3608 boost converter** stepping 7.4V → 5V (already purchased, 10-pack on hand)

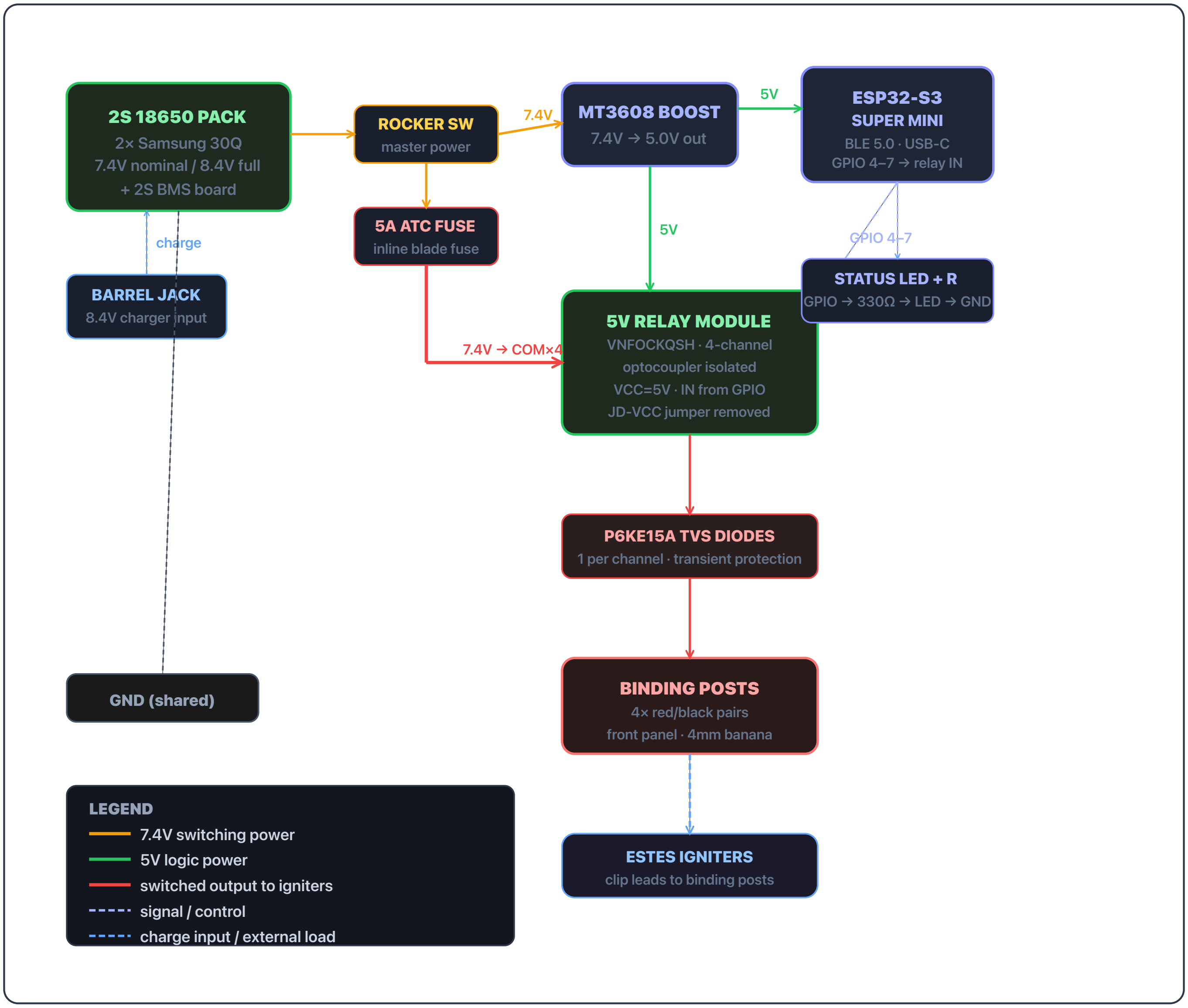
WAS Switching output designed for 12V external devices

NOW **7.4V switched output** to igniter binding posts — sufficient for Estes and hobby e-matches (fire at 0.5–1A), voltage floats between ~7V (depleted) and ~8.4V (full charge)

WAS 14.6V LiFePO4 charger (alligator clips)

NOW **8.4V wall charger** with 5.5×2.1mm barrel jack → matches DaierTek panel-mount barrel jacks already purchased

UPDATED SCHEMATIC



PARTS STATUS

Item	Status	Notes
2× Samsung 30Q 18650	✓ Have	Use 2 cells, set 2 aside as spares
DIANN single cell holders	✓ Have	Wire 2 in series
MT3608 boost converter	✓ Have	Set to 5.0V before connecting load
VNFOCKQSH 5V relay module (blue)	✓ Have	Remove JD-VCC jumper
ESP32-S3 Super Mini	✓ Have	—
P6KE15A TVS diodes	✓ Have	1 per output channel
ATC blade fuse + holder	✓ Have	Use 5A fuse

Glarks 4mm banana posts	✓ Have	4 red + 4 black
DaierTek barrel jacks	✓ Have	5.5×2.1mm panel mount
Nilight rocker switch	✓ Have	Master power
18 AWG wire	✓ Have	Power rails
Hammond 1455N1601BK	✓ Shipped	Panels in transit from FPE
2S BMS board (8A)	⚡ Order	Search "DIANN 2S BMS" ~$6
8.4V wall charger	⚡ Order	Search "iCreatin 8.4V charger" ~$10, 5.5×2.1mm barrel
NOYITO 24V relay (red board)	↩️ Return	Before July 10
DC HOUSE LiFePO4 battery	↩️ Return	Before July 10 · ~$24 back
THEPRO 14.6V LiFePO4 charger	↩️ Return	Before July 10 · ~$16 back

KEY WIRING NOTES

NOTE **MT3608 setup:** Connect battery pack to MT3608 input, measure output with multimeter, adjust pot to exactly 5.0V *before* connecting ESP32 or relay board.

NOTE **JD-VCC jumper:** Remove it from the relay module. This isolates the relay coil power from the optocoupler logic side.

NOTE **Switched output voltage:** The igniter terminals will see ~7–8.4V depending on battery charge state. Estes igniters fire reliably at this voltage.

NOTE **Cell holders in series:** Connect B+ of cell 1 to B- of cell 2. Pack B+ and pack B- go to BMS B+ and B- respectively. Load and charger connect to BMS P+ and P-.